

Heat Exchanger Network (HEN)

SYLLABUS for HEN

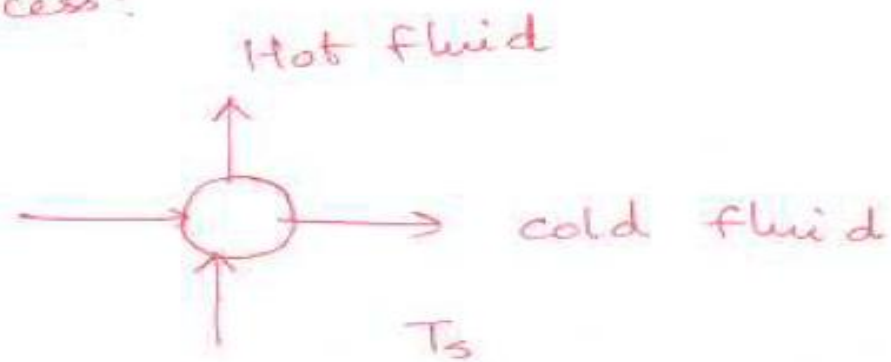
- Composite Curves
- Problem Table Algorithm
- Grand Composite Curve
- HEN Representation
- HEN Design for Maximum recovery
- Stream splitting

HEN Composite Curve

HEAT EXCHANGER NETWORKS

Composite Curves :

Let us take a single heat exchanger : Two stream recovery process:



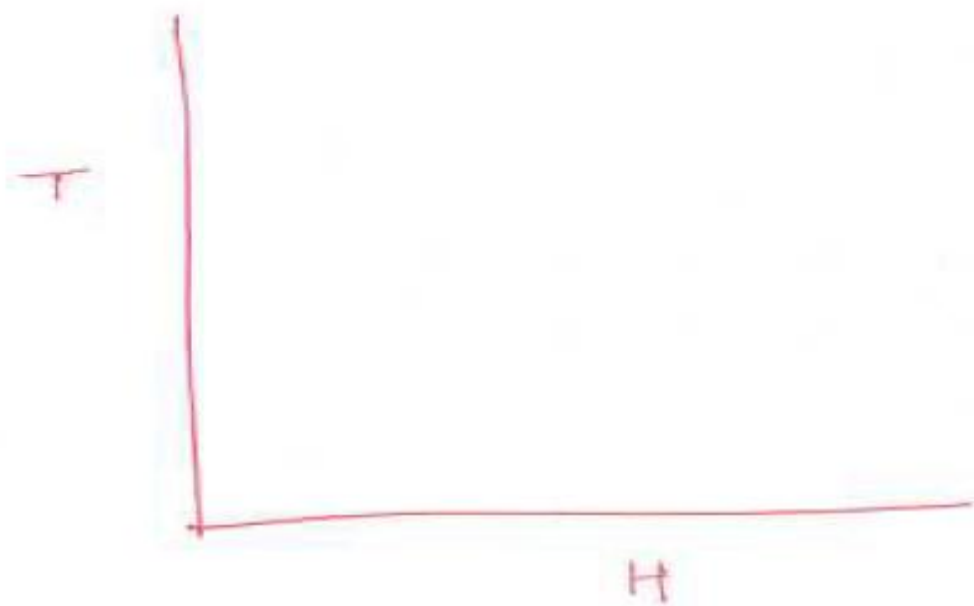
	T_s	T_r	$m C_p$
Cold	30	100	$0.2 \text{ MW/}^\circ\text{C}$
Hot	150	30	$0.1 \text{ MW/}^\circ\text{C}$

We assume ~~no~~ ^{some} temp. difference between them.

Now $\Delta H_{\text{cold}} = (100 - 30) = 70 \times 0.2 = 14 \text{ MW}$

$\Delta H_{\text{hot}} = -(150 - 30) \cdot 0.1 = -12 \text{ MW}$

If we plot ~~them~~ ΔT vs ΔH on a graph, if Origin is zero



$$\overset{\text{Inlet}}{H_{\text{cold}}} = 0.2 \times 30 = 6 \text{ MW/}^{\circ}\text{C}$$

$$\overset{\text{outlet}}{H_{\text{cold}}} = 0.2 \times 100 = 20 \text{ MW/}^{\circ}\text{C}$$

$$\text{|| by } \overset{\text{Inlet}}{H_{\text{hot}}} = 3 \text{ MW/}^{\circ}\text{C}$$

$$\overset{\text{outlet}}{H_{\text{hot}}} = 15 \text{ MW}$$

Let us put the coordinates together

Cold

$$(T_1, H_1) = (30, 6)$$

$$(T_2, H_2) = (100, 20)$$

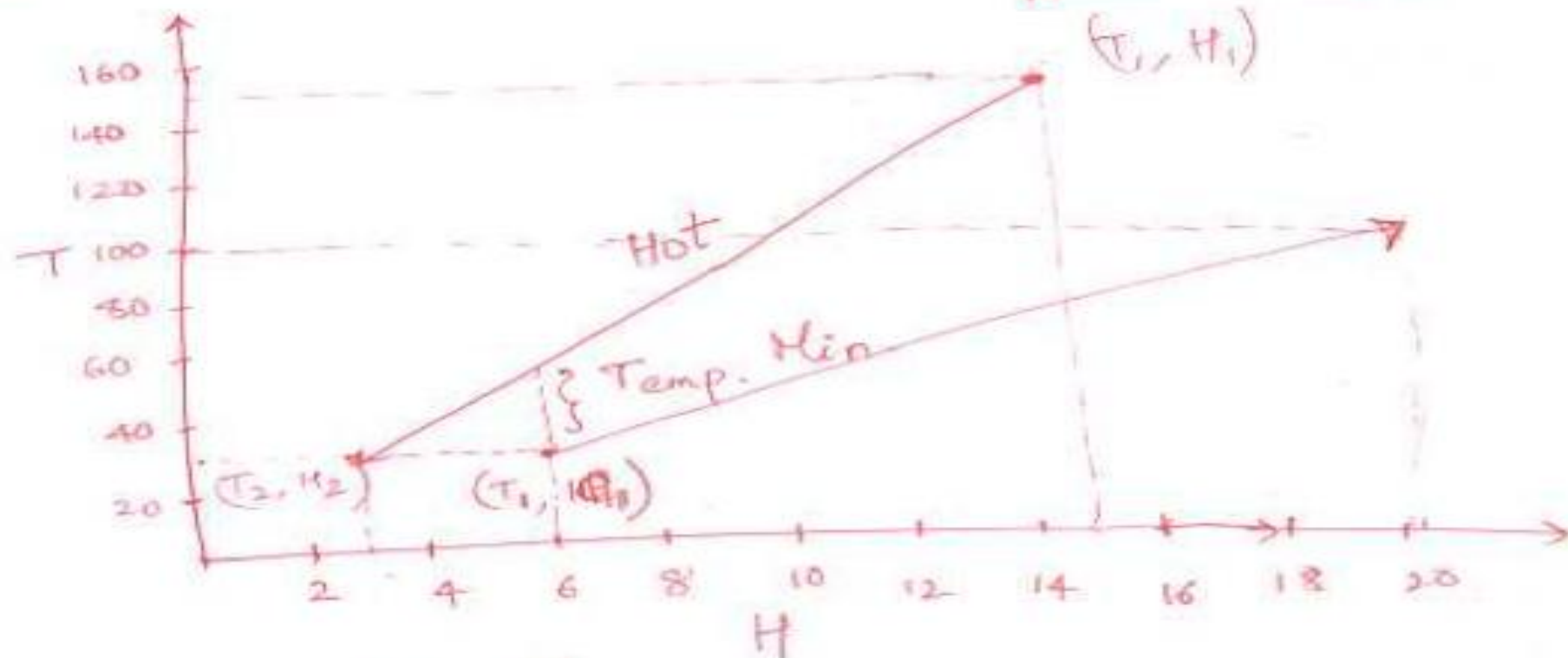
Hot

$$(T_1, H_1) = (150, 15)$$

$$(T_2, H_2) = (30, 3)$$

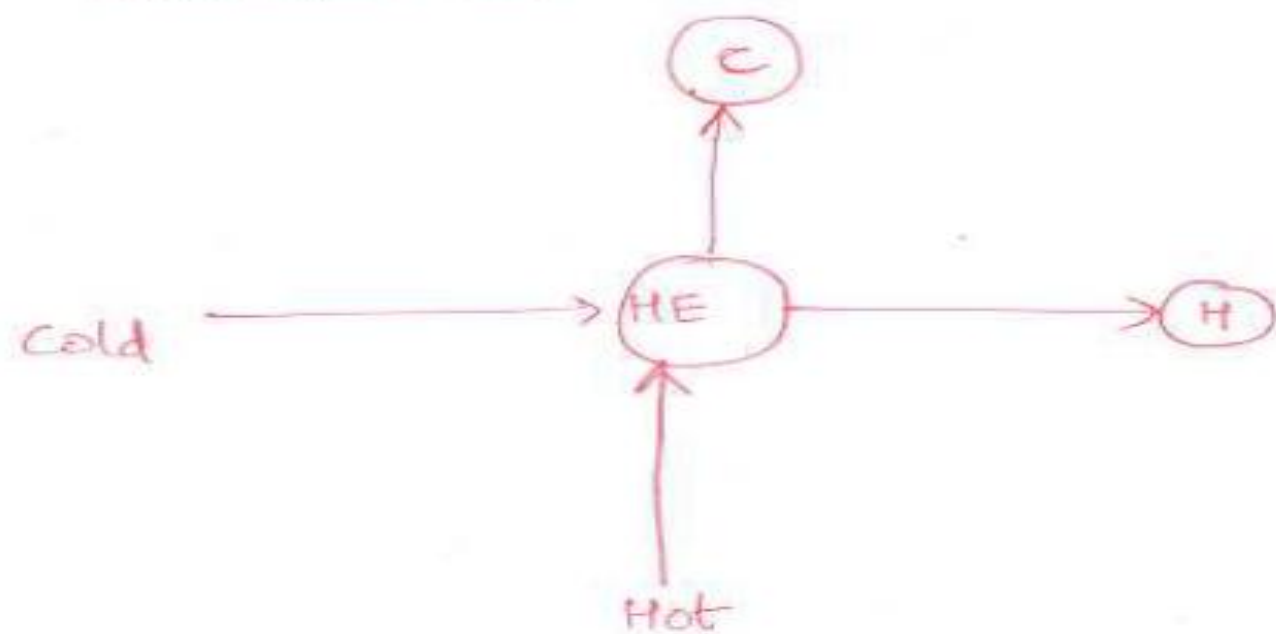
Need a heater

Plot them



Out of
Range

So therefore we need to get a ~~stream of~~ structure as



We need a heater and cooler to bring streams out of the range specified.

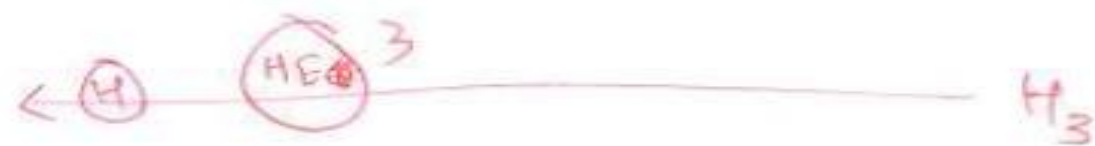
5 MW = Heat Demand on Cooler

3 MW = Heat Demand on Heater

Also Temp. Min. ~~cannot be~~ is at the starting point.



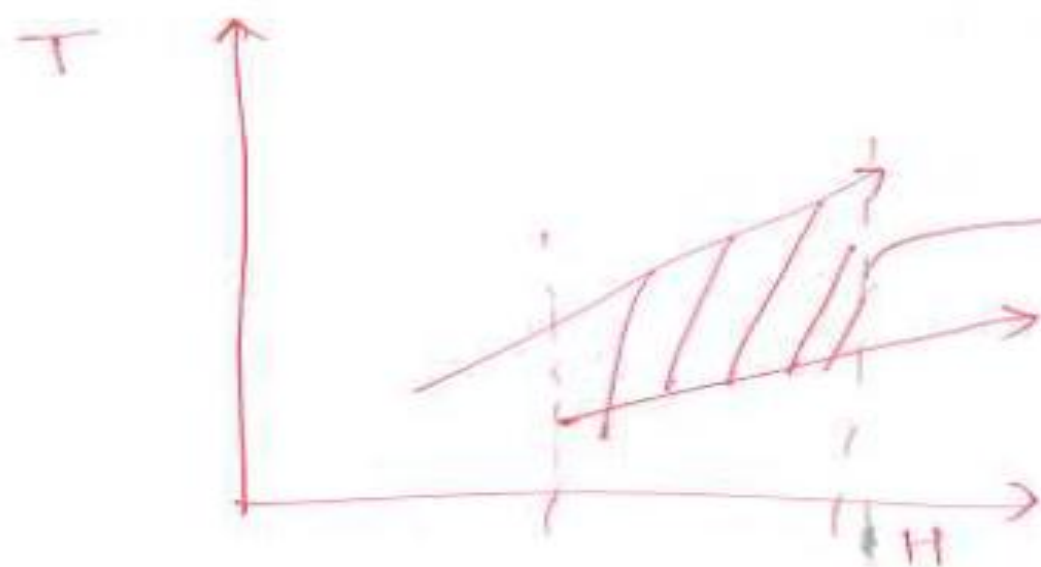
3 Heat Exchangers



Looks Awkward. We need ~~cost~~³ Heat Exchanger,
~~for~~³ heaters and ~~cooler~~³ for each stream. Too much cost.

What's gone wrong here?

recall the graph



~~HE~~ Heat extracted
for two streams
in one HE

The area shaded refers to the heat extracted
from hot stream to ~~cool down~~ heat the cold stream.

Therefore, we also need to see how we can develop composite curves for Multiple HES.

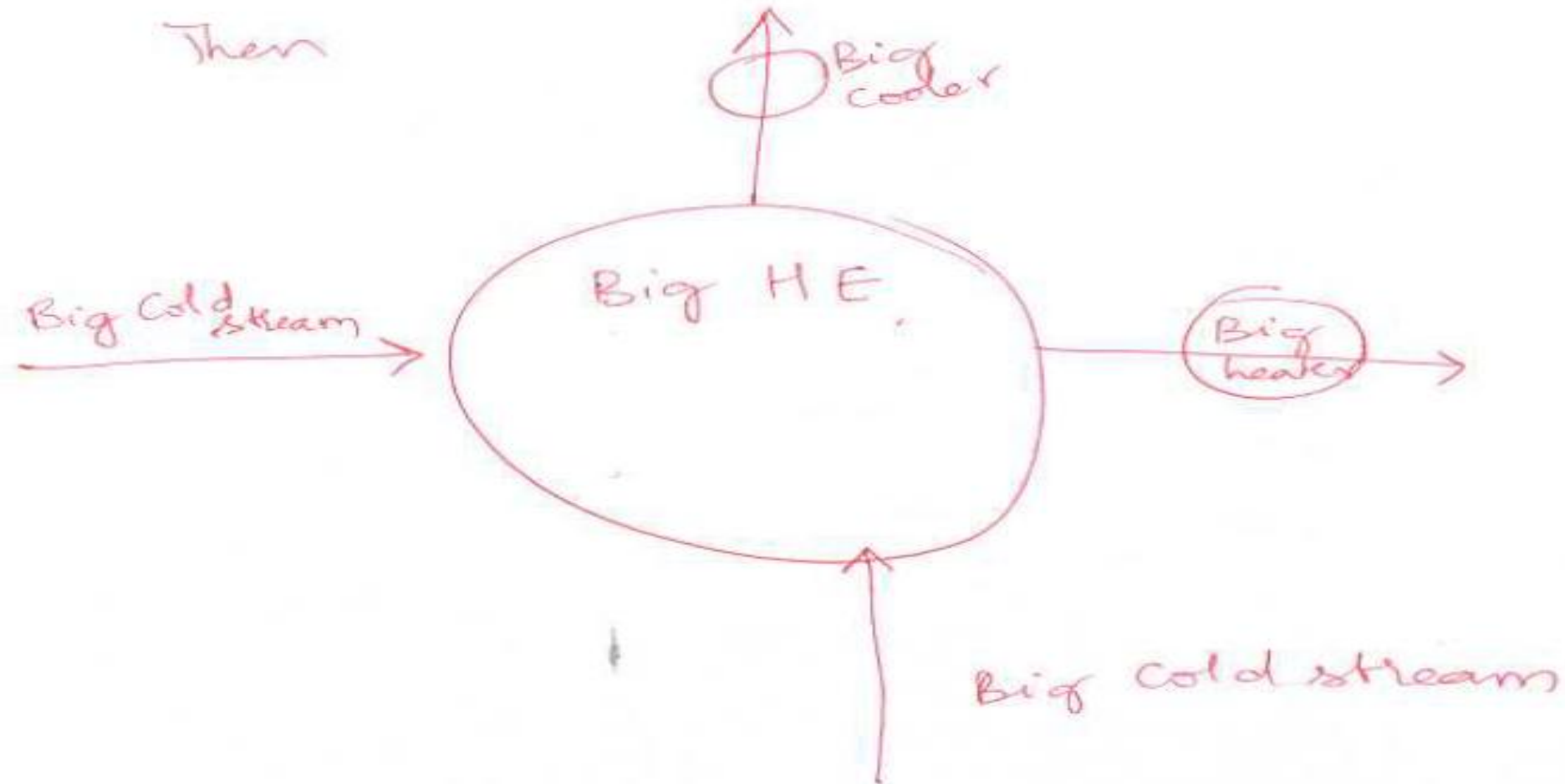
In fact for this problem, 1 heater and 1 Cooler can be sufficient to accommodate entire heat demand for all.
How?

Simple:

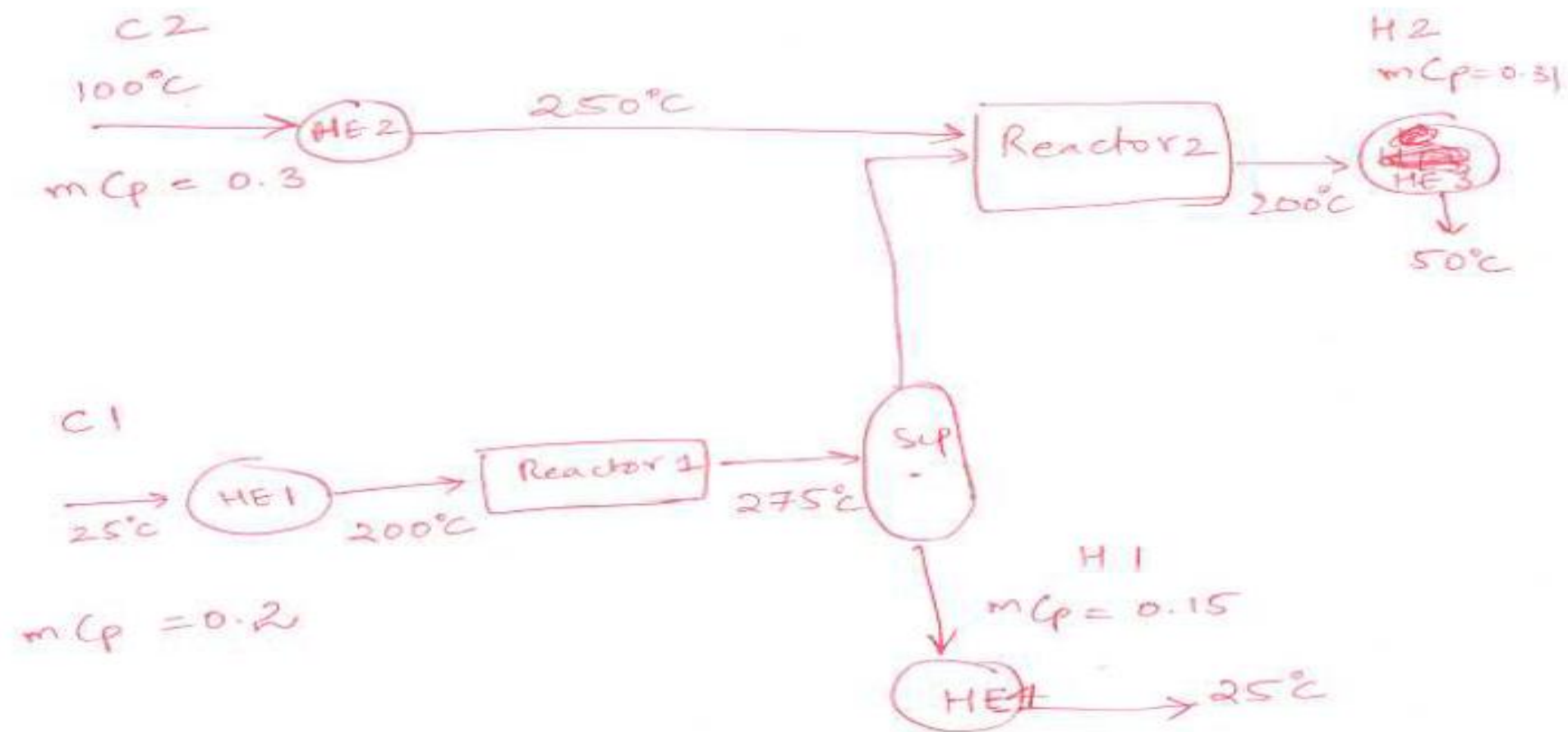
Visualise all cold streams to be One Cold stream with variable C_p (As streams get added or removed in due course of time)

Visualise all ~~too~~ hot streams to be one hot stream with variable Cp (As streams get added or removed).

Then



Example:



Summary:

	T_s	T_{target}	$m C_p$
C1	25	200	0.2
C2	100	250	0.3
H1	275	25	0.15
H2	200	50	0.31

Now let us see how we prepare two
Composite curves.

Then

We have streams of following Order (Stream Add/Deletion)
 ΣmC_p

Cold streams

25-100°C

C1 only

ΣmC_p

0.2

100-200°C

C1 + C2

$0.2 + 0.3 = 0.5$

200-250

C2 only

0.3

Hot streams

25-50°C

H1 only

ΣmC_p

0.15

50-200°C

H1 and H2

$0.15 + 0.31 = 0.46$

~~25-200°C~~
200-275°C

H2 only

~~0.46 + 0.31~~
0.15

Next step.

Calculate Enthalpies in this range.:

$$\sum mC_p \Delta T$$

$$3.75$$

25-50°C

50-200°C

$$0.46 (150)$$

$$~~72.75~~ \\ = 69$$

200-275°C

$$~~0.46~~ 0.15 \times 75 \\ = 11.25$$

Hot streams
Now Add them
Cumulative.

$$3.75$$

$$72.75$$

$$~~84~~ 84$$

Similarly for ~~hot~~^{cold} streams we have

25-100°C

$$\sum mC_p \Delta T$$

Cumulative

15

100-200°C

$$75 \times 0.2 = 15$$

65

200-250°C

$$100 \times 0.5 = 50$$

$$50 \times 0.3 = \frac{30}{15}$$

~~65~~

80

Total Enthalpy change for heat streams = 84 MW
Total Enthalpy change for cold streams = 80 MW

$$\cancel{84} \quad 84 > 80$$

Therefore we need a cooler to handle this problem.

If we plot the T, H curve we get a Composite Curve for our system. For that we need to enter data this way:

T (at which enthalpy varies)

Enthalpy

Hot streams :

25

0

50

3.75

200

72.75

275

84.

Cold streams

25

0

100

15

200

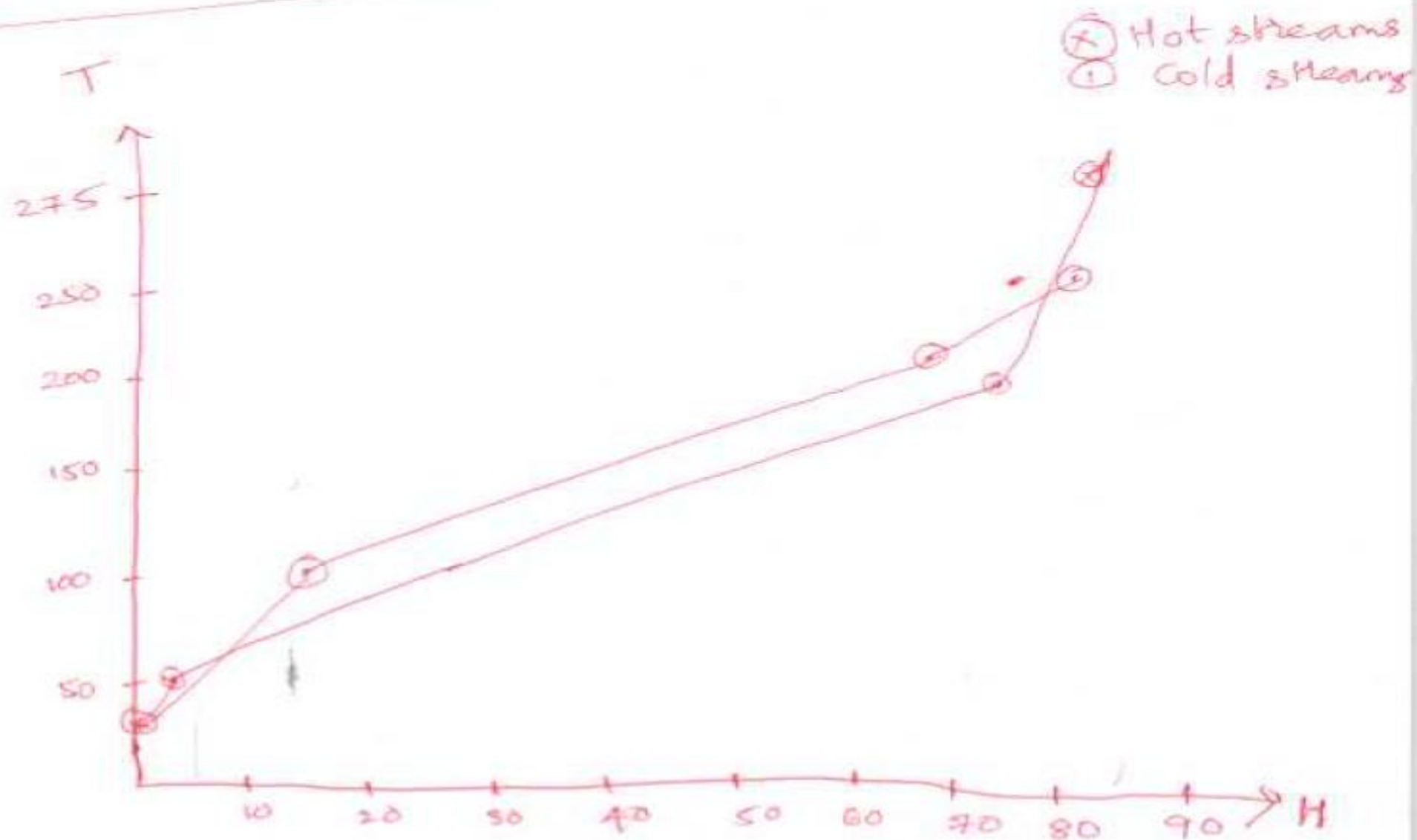
65

250

80.

Let us plot them together and see how our

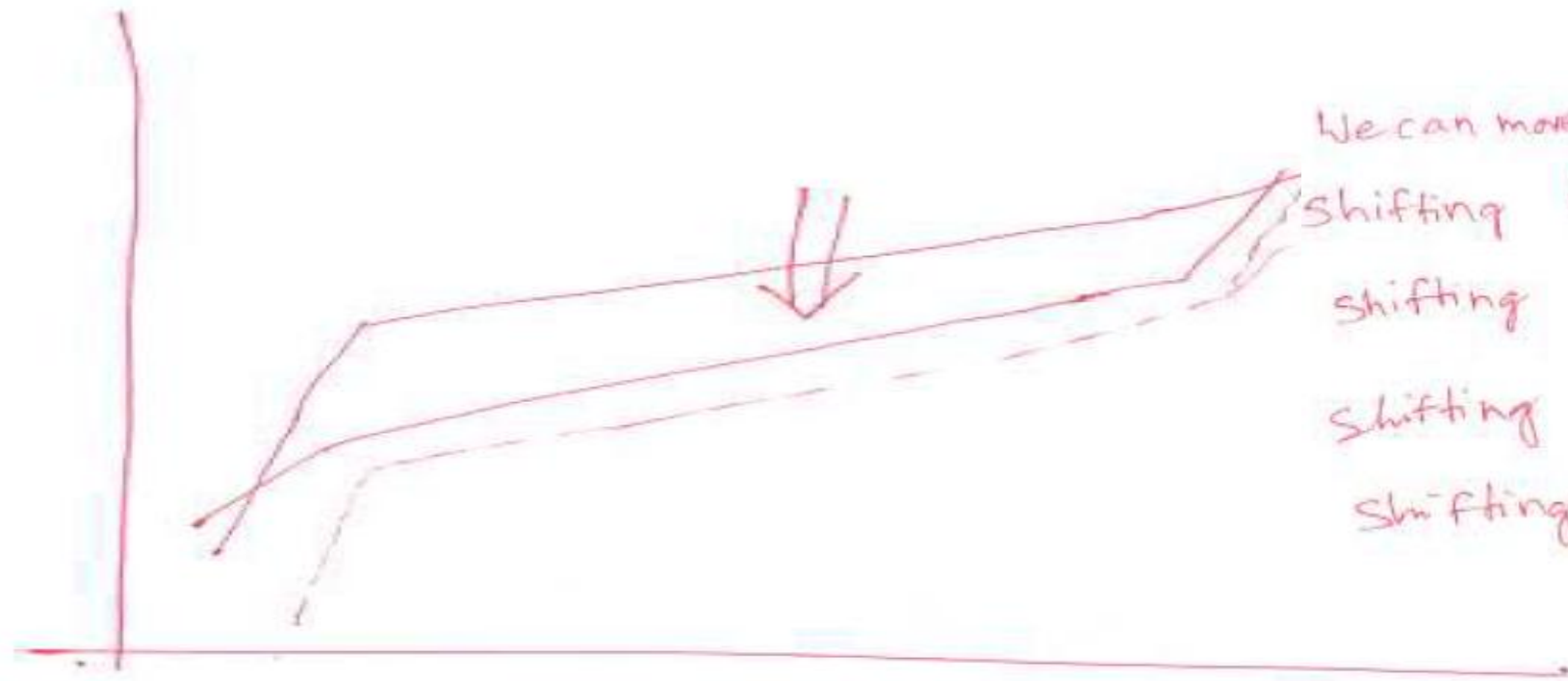
Composite looks like!



The ~~hot~~ hot composite curve is below
cold composite curve

Design Infeasible.

Therefore we have to shift the cold composite curve below
hot composite curve. What will happen



We can move in many ways.

Shifting hot curve to the left

Shifting cold curve to the right

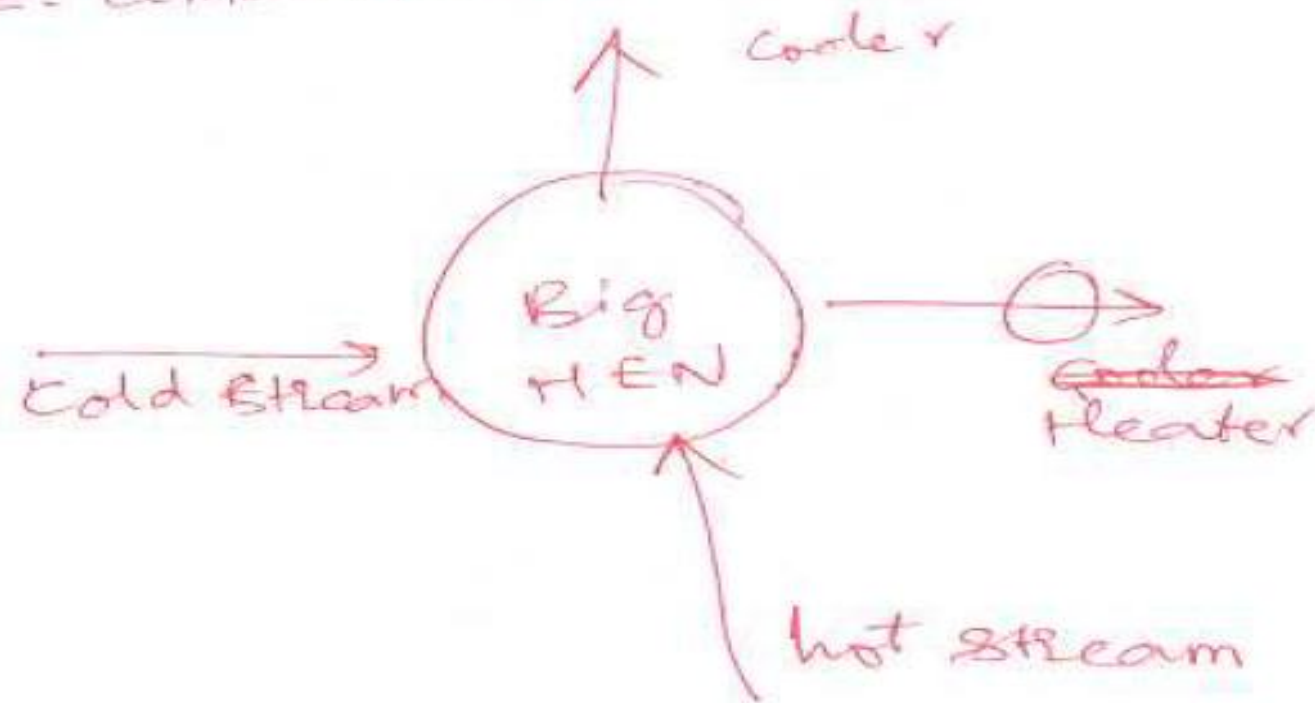
Shifting hot curve above

Shifting cold curve below.

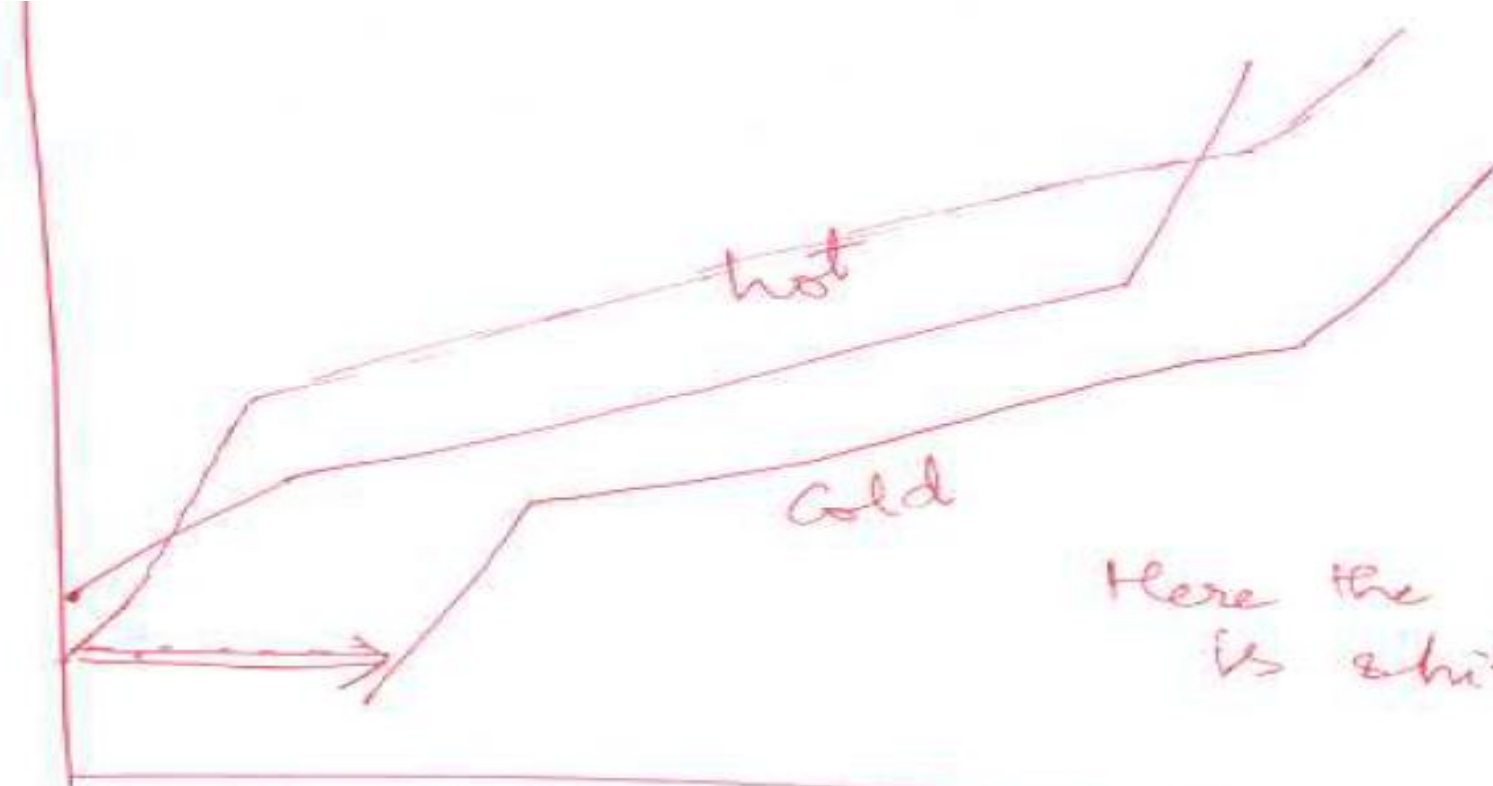
What can be done now?

Usually cold curve is shifted such that there exists a heating and cooling duties for the ~~component~~ combination of hot & cold composites.

Recall



For this case, the cold stream is shifted as follows:



Here the cold stream is shifted right

Gold stream:

Starting point

~~25 MW~~

$$(T, H) = \text{25} (25, 20)$$

$$(100, 435)$$

$$(200, 85)$$

$$(\underline{250, 100})$$

If ~~it~~ stream 1 Enthalpy is increased to 20 MW,
what does it mean?

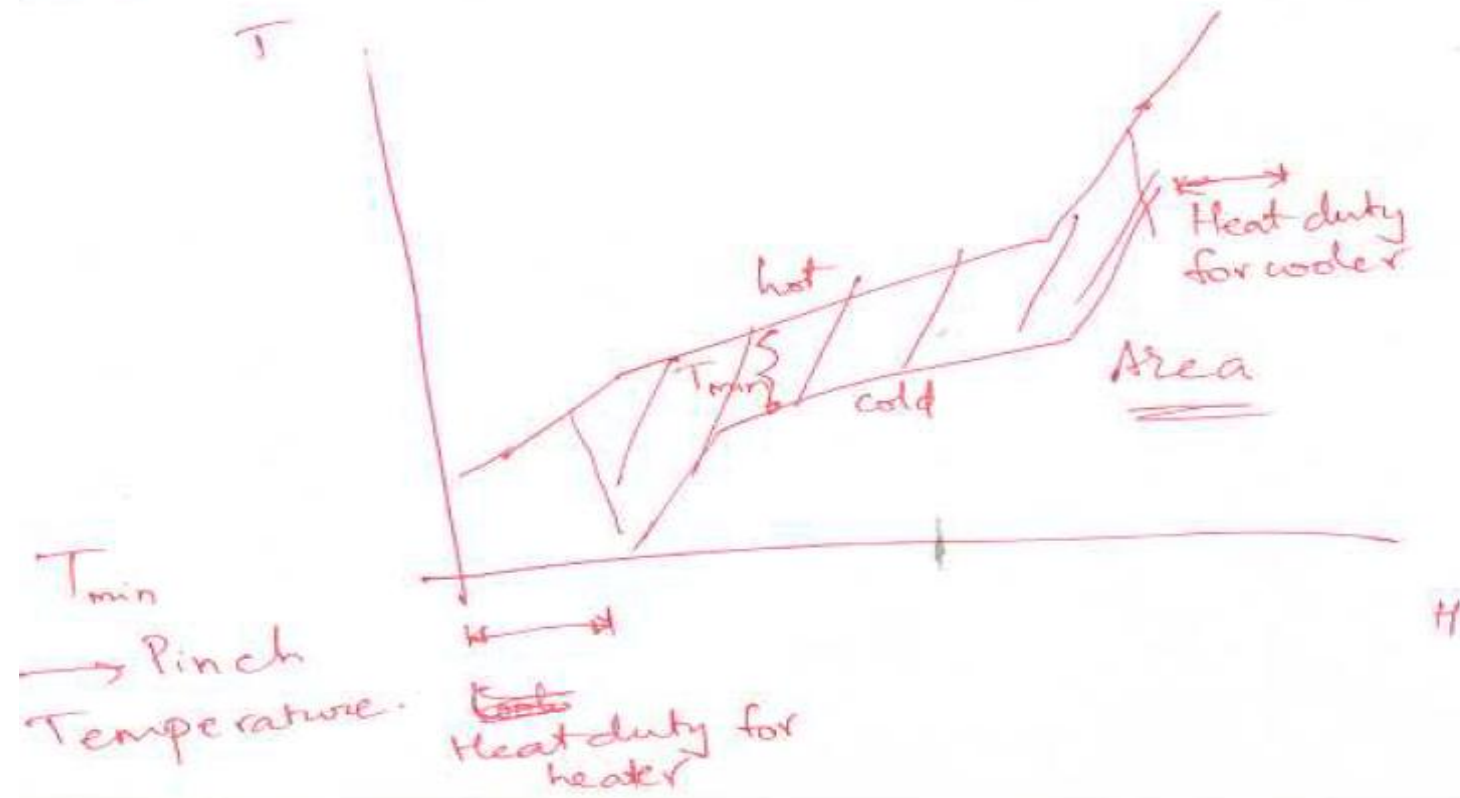
$$20 = \overset{\text{Same}}{\cancel{m}} \underbrace{(C_p)}_{\substack{\downarrow \\ \text{CP values are } \cancel{\text{adjusted}} \text{ How?} \\ \text{or } \Delta T \text{ changed.}}} (\Delta T)_{\text{increased}}$$

CP values can be increased by more throughput?
But then $\Delta T = 0$ (since Ref Temp is @ 25°C)
So Its Ref Temp. that has to adjusted +
~~m~~ C_p for C_1 has to be increased
such that we have a shift of 20 MW.

But then Ref. value of hot stream also get
adjusted and combined effect nullifies things.
stream temp.

again.
Therefore it is best to choose our stream Temp.
in such away that we ~~have~~ donot encounter
this problem

Back to the Composite again:



In this example, we now see how shift will change things

Shift	ΔT_{min}	Cooling Utility (MW)	Heating Utility (MW)
12	~0	12	8
16.35	10	16.35	12.35
20	~15	20	16
20.95	20	20.95 20.95	16.95

Note that the difference between Cooling Utility and Heating Utility does not change.

Cooling & Heating utilities technically called energy targets.